Supervised Machine Learning Approach

Without AI

1. Student Attendance Prediction

Data Collection

Historical records of overall attendance data, as well as focused attendance data from individual students.

Availability of labelled data

Available to faculty

Utility of the proposed solution

Supports specific attendance challenges of students, and will help course instructors personalise their approach based on how likely a student is to attend classes.

Challenges

Might introduce bias if data is not properly sanitized, and outputs might lead to erroneous conclusions.

2. Marks Prediction

Data Collection

Historical overall marks data, as well as student-specific data per subject.

Availability of labelled data

Freely available

Utility of the proposed solution

Personalised approach to students' academic performances.

Challenges

Predicted marks in any subject might lead to unforeseen consequences during actual evaluations.

3. Classroom Availability Prediction

Data Collection

Get timetable data from different departments at a block, and overall classroom availability from block in-charge.

Availability of labelled data

Not easy to get data, needs extensive coordination between multiple stakeholders.

Utility of the proposed solution

Will help course designers and department admin allocate classes based on when they might be free over a certain period of time, also useful for planning events.

Challenges

Data collection is a significant issue, as well as last-minute changes in schedule due to circumstances which cannot be reasonably predicted.

With AI

1. Student Placement Prediction

Problem Statement

Predict whether a student is likely to be placed during campus recruitment based on academic and extracurricular performance.

1. Data Collection

Data can be collected from university records and placement cell databases, including:

- GPA/CGPA
- Attendance percentage
- Internal Assessment (CIA) marks
- Internship experience
- Certifications
- Programming skills
- Projects completed
- Soft skills assessment
- Aptitude test scores
- Placement status of previous batches

2. Availability of Labelled Data

Yes.

Historical placement records already contain labels such as:

- Placed
- Not Placed

or

- Company Name
- Salary Package

These serve as target variables for supervised learning.

3. Utility of the Proposed Solution

- Helps identify students requiring placement training.
- Assists the placement cell in planning training programs.
- Improves campus placement success rate.
- Provides personalized career guidance.

4. Challenges

- Missing or incomplete student records.
- Changing hiring criteria across companies.
- Class imbalance if most students are placed or unplaced.
- Privacy concerns while handling student information.

2. Student Academic Performance Prediction

Problem Statement

Predict the final GPA or semester performance of a student using continuous assessment and academic activities.

1. Data Collection

Possible features include:

- CIA scores
- Previous semester GPA

- Attendance percentage
- Assignment marks
- Lab performance
- Quiz scores
- Participation in class
- Number of backlogs
- Study hours (survey)

2. Availability of Labelled Data

Yes.

Historical academic records provide labels such as:

- Final GPA
- Semester percentage
- Grade obtained

These numerical values are suitable for supervised regression models.

3. Utility of the Proposed Solution

- Identifies academically at-risk students early.
- Enables faculty to provide remedial support.
- Improves academic planning.
- Enhances overall student performance.

4. Challenges

- Student motivation is difficult to quantify.
- External factors affecting performance may not be available.
- Missing assessment records.
- Data imbalance across departments.

3. Student Attendance Risk Prediction

Problem Statement

Predict whether a student is likely to fall below the minimum attendance requirement before the end of the semester.

1. Data Collection

Relevant features include:

- Daily attendance records
- Previous attendance history
- Number of leaves taken
- Timetable
- Medical leave records
- Semester workload
- Participation in extracurricular activities

2. Availability of Labelled Data

Yes.

Existing attendance records can be labelled as:

- Attendance Shortage (Yes/No)

or

- Eligible for Examination
- Not Eligible

These labels are already maintained by the university.

3. Utility of the Proposed Solution

- Generates early alerts for students.
- Helps faculty counsel students.
- Reduces attendance shortages.
- Improves compliance with university attendance regulations.

4. Challenges

- Sudden medical emergencies affecting attendance.
- Late attendance updates.
- Legitimate absences may not be reflected immediately.
- Maintaining accurate daily attendance records.

<u>Comparison</u>

Aspect	**Without AI**	**With AI (Generated Analysis)**
Problems Identified	Student Attendance Prediction, Marks Prediction, Classroom Availability Prediction	Student Placement Prediction, Student Academic Performance Prediction, Student Attendance Risk Prediction
Relevance to CHRIST University	Highly contextual and institution-specific. Includes operational problems faced by departments and administration.	Focuses on common academic applications of supervised learning that are applicable to CHRIST University but are more generic.
Data Collection	Briefly identifies the main data sources (attendance records, marks, timetable data).	Provides a detailed list of potential features (GPA, attendance, internships, assignments, CIA scores, certifications, etc.) for each problem.
Availability of Labelled Data	Short and practical observations (e.g., available to faculty, freely available, difficult to obtain).	Explains why labelled data already exists (historical records, placement status, GPA, attendance shortage

		labels) and how it is suitable for supervised learning.
Utility of the Solution	Focuses on practical institutional benefits such as personalized teaching, academic support, and classroom scheduling.	Provides broader benefits including early intervention, administrative planning, placement preparation, academic counseling, and improved student outcomes.
Challenges	Mentions realistic institutional concerns such as biased data, inaccurate predictions, and difficulties in data collection or schedule changes.	Discusses technical and machine learning challenges including missing data, class imbalance, privacy, changing recruitment trends, and unpredictable external factors.
Level of Detail	Concise and suitable for a short assignment or classroom submission.	More comprehensive and descriptive, suitable for reports or detailed assignments.
Machine Learning Perspective	Focuses primarily on the application problem rather than the ML methodology.	Clearly frames each problem as a supervised learning task (classification or regression) and identifies the target labels.
Practicality	Includes Classroom Availability Prediction, which is a practical administrative use case but may require significant coordination for data collection.	Includes Student Placement Prediction, which is a widely accepted supervised learning application with readily available labelled historical data.
Innovation	Classroom Availability Prediction is relatively	Student Placement Prediction is a standard educational analytics

	unique and institution-specific.	problem commonly discussed in machine learning applications.
Ease of Implementation	Attendance and Marks Prediction are relatively straightforward. Classroom Availability Prediction is more difficult due to fragmented data sources.	Attendance and Academic Performance Prediction are relatively straightforward. Placement Prediction requires integration with placement records but is feasible.
Overall Strength	Strong practical understanding of institutional problems with concise explanations.	Stronger technical explanation of supervised learning concepts with richer justification and machine learning terminology.